Long-Term Projects

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In this document, I will describe some long-term projects which I am thinking about and intend to "complete" sometime in the future. The projects here mostly have an operator-algebraic flavor, though there are plenty of interactions with other aspects of mathematics. More importantly, each project contains a reasonable mix of a concrete problem (or set of problems) and some theory-building. We have included a section on updates (see Section 4) which is meant to detail the progress made towards each project.

1 Semiprojectivity of $C^\star$-Algebras & Noncommutative Shape Theory

In NCG, it is common to take inspiration from classical (algebraic) topology, geometry and analysis to motivate the introduction of interesting new structures which capture old and new phenomena. Classical shape theory was developed by Borsuk as a way of extending results from the homotopy theory of CW complexes to compact metrizable spaces; see [5] for a survey. In [2], Blackadar introduced shape theory for $C^\star$-algebras and, in particular, introduced semiprojective $C^\star$-algebras as the noncommutataive analog of spaces which satisfy the homotopy extension property.

Definition 1.1. [2]

Let A be a separable $C^\star$-algebra.

1. A is semiprojective iff for any $B \in \mathrm{C^\star Alg}$, any increasing sequence $(J_n)_{n\in\mathbb{N}}$ of closed ideals in B with J being the closed ideal generated by the sequence and any $\star$-homomorphism $\phi : A \to B/J$, ϕ can be lifted to a $\star$-homomorphism $\psi : A \to B/J_n$ for sufficiently large n.
2. A is approximately semiprojective iff it is isomorphic to a sequential inductive limit of semiprojective $C^\star$-algebras.

In the literature, an approximately semiprojective $C^\star$-algebra is often also said to have a strong shape system.

A major conjecture about approximately semiprojective $C^\star$-algebras is the following.

Conjecture 1.2 (B. Blackadar, 1985, [2, 4.4, p.266])**.** Every separable $C^\star$-algebra is approximately semiprojective.

In my master's thesis, I managed to construct a potential obstruction to this conjecture that we would like to investigate in the future. To be precise, I introduced a new class of $C^\star$-algebras, known as the Schochet $C^\star$-algebras. These are exactly those $C^\star$-algebras which satisfy the homotopy lifting property with respect to the Calkin projection on an infinite-dimensional, separable $\mathbb{C}$-Hilbert space. More precisely, the obstruction I showed is the following result.

Proposition 1.3. A. Vats, 2026

Every approximately semiprojective $C^\star$-algebra is approximately Schochet.

So, one might hope to construct a counterexample to Conjecture 1.2 by finding an approximately Schochet $C^\star$-algebra which is not approximately semiprojective. This involves a study of lifting properties which is more nuanced than what I have conducted in my master's thesis and some of that intersects with Prof. Dor-On's work. For instance, he (and his collaborators) recently proved, in [4, Theorem 1.2, p.2], a quantitative version of a theorem of Eliers, Loring and Pedersen which provides quantitative bounds for matricial semiprojectivity when a winding number obstruction vanishes. This theorem was motivated also by an old question of (Paul) Halmos on almost commuting matrices and the proof techniques utilized do have close relations to the way (approximate) semiprojectivity is analyzed for concrete $C^\star$-algebras. Moreover, it is quite likely that they can be suitably adapted to dealing with path lifting problems, which are useful for analyzing Schochet $C^\star$-algebras and could be the way to construct a counterexample to Conjecture 1.2.

2 Homotopical/Categorical Approach to Classifying Leavitt Path Algebras

One of my goals during the PhD was to work further on applying tools from (higher) algebraic K-theory, which rely heavily on modern homotopy theory, to operator algebras. A new direction in this vein which was introduced to me by Prof. Dor-On was in studying graded classification conjectures for Leavitt path algebras. These conjectures actually have their origins in the decidability problem for conjugacy of subshifts of finite type (SFTs) (see [7]).

Conjecture 2.1 (Hazrat's Graded Morita Equivalence Classification Conjecture, [1, Conjecture 1.3, p.2])**.** Let A and B be two finite essential matrices. The following are equivalent:

1. A and B are shift equivalent.
2. The Cuntz-Krieger graph $C^\star$-algebras $\mathcal{O}_A$ and $\mathcal{O}_B$ are stably equivariantly homotopy equivalent.

3. The Leavitt path algebras L_A and L_B are graded Morita equivalent.

The next theorem indicates that the conjecture has a positive answer in a homotopical reformulation.

Theorem 2.2. B. Bilich, A. Dor-On, E. Ruiz, 2024, [1, Theorem A, p.2]

Let A, B be essential square matrices with entries in $\mathbb{N}$. The following are equivalent:

1. A and B are shift equivalent.
2. $\mathcal{O}_A$ and $\mathcal{O}_B$ are stably equivariantly homotopy equivalent.

The aforementioned reference serves as the first paper which uses bicategorical techniques to classify graph $C^\star$-algebras up to graded stable homotopy equivalence & it is the first piece of evidence that the systematic use of homotopy theory might provide a solution to Conjecture 2.1. Our interest also extends towards finding a possible counterexample to this conjecture. Indeed, in [8], it is shown that the Williams Conjecture, which says that shift equivalence is the same as strong shift equivalence, is false for irreducible SFTs and this is done by introducing the sign-gyration invariant for conjugacies between shifts, as well as a counterexample of two 7×7 matrices satisfying the right selection of properties so as to form a counterexample. This leads us to a natural question that we hope to address during the PhD.

Question 2.3. Is there an algebraic invariant that we can use to dstinguish between the Leavitt path algebras of the 7×7 matrices A and B introduced by Kim & Roush in [8, p.554]?

One reason for believing that this question can be attacked via K-theoretic methods is via [9] and [11], where a relationship between the Kim-Roush sign-gyration invariant and algebraic K_2. If we can show that Wagoner's K_2 is an invariant for graded isomorphism of Leavitt path algebras; this would show that the counterexample of Kim and Roush is also a counterexample to Conjecture 2.1. In general, a more systematic understanding of the relationship between these invariants would lead us towards better results in relation to the aforementioned conjecture. Another major reason for pursuing this line of inquiry is due to the following theorem.

Theorem 2.4. [3, Theorem 8.8.1, p.25]

Let E and F be finite regular graphs. The following

1. The Leavitt path algebras $L(E)$ and $L(F)$ are graded Morita equivalent.
2. The singularity categories $\mathbf{D}_{\mathrm{sng}}(A(E))$ and $\mathbf{D}_{\mathrm{sng}}(A(F))$ are triangulated equivalent.

The singularity categories are constructed by taking quotients of bounded derived categories & such categories are best understood from the viewpoint of ∞-category theory. One possible way to study statement (2), for example, would be to construct the derived ∞-categories associated to the two Leavitt path algebras and compare their structures first. My background in homotopy theory will prove useful in working out the details of this promising approach.

3 Computations of K-Theory of Crossed Products of Hecke $C^\star$-Algebras

The project in Section 2 focuses more on computations of algebraic K-theory and its interactions with homotopy theory. However, there is much work to be done on the topological front as well.

Question 3.1. Let G be a locally compact Hausdorff group and let A be a $C^\star$-algebra which is a G-dynamical system . We can form a $C^\star$-algebra $A \times_r G$, which we refer to as the reduced crossed product of A with G. What is $\mathrm{K}_i\left(A \times_r G\right)$ when $i \in \{0, 1\}$?

In general, this is a very difficult question to answer, though the Baum-Connes Conjecture provides an answer to this problem for many groups through reduction to an equivariant KK-theory computation. However, the difficulty of this is already apparent when investigating crossed products by finite groups. The goal of this project is to use extensions of Hecke algebras by finite groups as a testing field of a homotopical persspective towards Question 3.1. To be precise, we are interested in studying the K-theory of Hecke $C^\star$-algebras of right-angled Coxeter groups.

Definition 3.2

A Coxeter system (W, S) consists of group W freely generated by a finite set S with relations:

$$(st)^{m_{st}} = 1_W,$$

for exponents $m_{st} \in \mathbb{N} \cup \{\infty\}$ so that:

1. $\forall s \in S : m_{s,s} = 1,$
2. $\forall s, t \in S : s \neq t \Rightarrow m_{st} \geq 2,$
3. $\forall s, t \in S : m_{st} = m_{ts}.$
4. If $m_{st} = \infty$, then no relation is imposed on s and t.

(W, S) is right-angled iff $m_{s,t} \in \{2, \infty\}$ for all distinct $s, t \in S$.

These groups find their easiest motivation within geometry, particularly in the structure theory of finite reflection groups ([6]). Now, Hecke algebras are one-prameter deformations of the (complex) group algebra of a Coxeter group and have foundational applications in the theory of quantum groups & knot invariants. In our case, we are interested in Hecke $C^\star$-algebras arising from right-angled Coxeter groups.

Question 3.3. Let (W, S) be a right-angled Coxeter system and let $\vec{q}$ be a multiparameter on[a] S. Then, it is possible to define a $C^\star$-algebra $C_r^\star(W, S, \vec{q})$ from the triple $(W, S, \vec{q})$. If Γ is the Cayley graph of (W, S), then $\mathrm{Aut}(\Gamma)$ acts on $C_r^\star(W, S, \vec{q})$ so we can define $C_r^\star(W, S, \vec{q}) \times_r \mathrm{Aut}(\Gamma)$. Now, what is $\mathrm{K}_i\left(C_r^\star(W, S, \vec{q}) \times_r \mathrm{Aut}(\Gamma)\right)$ for $i \in \{0, 1\}$?

[a] This is just a map $S \to (0, 1]$.

The next theorem is a major result which indicates that the question above is worth investigating.

Theorem 3.4. S. Raum, A. Skalski, [10, Theorem 4.2, p.10-12]

Let (W, S) be a right-angled Coxeter system, let Γ be the associated commutation graph & let $\mathcal{C}$ be the set of all cliques (including the empty clique) of Γ. The following statements hold true:

1. The map $C \mapsto [p_C]$ induces an isomorphism $\mathbb{Z}^{\mathcal{C}} \simeq \mathrm{K}_0\left(C^\star(W)\right)$.
2. $\mathrm{K}_1\left(C^\star(W)\right) \simeq 0$.

$\mathrm{K}_0\left(C^\star(W)\right)$ is a free abelian group whose rank is the number of cliques in the commutation graph of (W, S) and it is generated by the projections from the copies $C^\star(\mathbb{Z}/2) \subseteq C^\star(W)$ associated with the Coxeter generators $s \in S$.

Much of the proof involves showing that a few key maps which are natural to the set-up are KK-equivalences, for the purpose of showing that a certain reduction map from the universal graph product to the reduced graph product is a KK-equivalence. An interesting observation that could lead to a resolution of Question 3.3 is the fact that a key gluing lemma used to prove Theorem 3.4 is known to be true in model categories or under even weaker conditions, such as those of a category of cofibrant objects. So, it might be interesting to consider finding a category of cofibrant objects on the category of $C^\star$-algebras with morphisms being completely positive maps, cofibrations being conditional expectations and weak equivalences being KK-equivalences. This would imply the key gluing lemma, both in the non-equivariant and equivariant case. My background will prove useful in investigating this approach.

4 Updates

References

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